Calibration Procedure

Objective

1. Have at least 23% (19% for SPIF21) of the total truck weight for SPIF23 on the front axle.
2. Share at least half of the tandem axle weight (front + rear tandem axle weight) on the lift axle.

Steps

1. Measure the axle spread (front to lift axle distance, lift to front tandem axle distance and tandem axle separation) to determine the schedule category (SPIF21 or SPIF23).
2. Determine the suspension type of the truck to be calibrated.
3. Connect the system to the laptop/PC with the appropriate cable and open the software.
4. Select the appropriate schedule information and suspension type in the software.
5. Weigh the truck empty with lift axle raised. Get the front axle, front tandem axle and rear tandem axle weight. The sum gives the total(gross) truck weight.
6. Note down these weights and the tandem sensor value.
7. Calculate one third of the tandem axle weight ((front tandem + rear tandem)/3) and try to achieve that weight on the lift axle by varying the pressure input.
8. After maintaining one third of the tandem weight on the lift axle, weigh the front axle with lift axle in the same position to confirm the front axle has at least 23% of the total truck weight for SPIF23 or 19% of the total truck weight for SPIF21 (objective 1).
9. If the weight on the front axle is less than what is necessary according to SPIF regulation, decrease the weight on the lift axle by lowering the pressure until the first objective is met.
10. Once the first objective is achieved, weigh tandem axles to confirm all the three axles; two tandem axles and the lift axle is sharing equal weight (objective 2).
11. Note down all weights, tandem sensor values, and ride bag pressure values for the lift axle in the lowered condition after both objectives have been met.
12. Repeat the entire process for loaded condition which will give another set of data for the truck under load.
13. Input the data into their respective field and hit calc button under each input field to calculate the gross weight of the truck. If everything was done correctly, the total weight of the truck with lift axle raised and lowered should be identical with ±200KG error.
14. Once the total weight of the truck is verified within the error limit under empty and loaded conditions, the lower, the lift and the MTP values can be calculated by pressing the calculate button under the related tab.
15. The software then spits out the values which then can be fed into the lower, the lift and the MTP input fields and pressing the send button would effectively calibrate the truck. We

can verify if the truck is calibrated by looking into the lift, the lower and the MTP values under ‘Sensor Readings’ tab.

Note

During calibration, quite often it may seem impossible to meet both objectives, which is indeed true. Of the two objectives, the first is the more critical, and we must make a trade-off—either striving to achieve both or ensuring the first is met while approaching the second as closely as possible.

For instance, if the front axle carries 35% of the total truck weight but the lift axle carries only 20% of the tandem weight, we can reduce the load on the front axle to help the lift axle approach 50% of the tandem weight. However, while reducing the front axle load, we must ensure that the first objective is not violated. If both conditions ultimately prove unattainable, we should meet the first objective at its allowable limit which will bring the second as close to the regulatory requirement as possible.